

Self-Fulfilling Fluctuations in HANK Economies[†]

By SUSHANT ACHARYA AND JESS BENHABIB*

We show that in heterogeneous agent New Keynesian (HANK) economies with countercyclical risk, the natural interest rate is endogenous and co-moves with output, leaving the economy susceptible to self-fulfilling fluctuations. Unlike in representative agent New Keynesian models, the Taylor principle is not sufficient to guarantee uniqueness of equilibrium in HANK if risk is even mildly countercyclical: Multiple bounded equilibria exist, no matter how strongly monetary policy responds to changes in inflation. For an active monetary policy to eliminate self-fulfilling fluctuations, it must stabilize the endogenous natural rate fluctuations. Alternatively, a passive monetary and active fiscal regime can also eliminate equilibrium multiplicity. (JEL E12, E23, E31, E32, E43, E52, E62)

Economists have long argued that shifts in expectations can generate fluctuations in output and inflation even absent any change in fundamentals, and that countercyclical stabilization policy plays an important role in mitigating such fluctuations (Keynes 1936). Left unchecked, these self-fulfilling beliefs can destabilize the economy. Consequently, a central tenet in the science of monetary policy is that when inflation rises, a central bank should increase nominal interest rates aggressively to raise the real interest rate. Without this aggressive response, a self-fulfilling inflationary process may take hold: Households' expectations of higher inflation lowers the real interest rate, which in turn stimulates demand and pushes up actual inflation, thereby confirming the initial belief. Raising nominal rates more than one-for-one with inflation nips these nonfundamental beliefs in the bud, stabilizing the economy. Arguably, the determination to keep inflation in check and inflation expectations anchored underlies the steep increase in policy rates around the world in response to the high inflation during the recovery from the COVID-19 recession.

The representative agent New Keynesian (RANK) models, which provide the theoretical backbone for raising nominal rates more than one-for-one with inflation—the Taylor principle—abstract from inequality, market incompleteness, and the distributional effects of monetary policy. In recent years, there has been growing interest among both academics and policymakers to understand how these

*Acharya: University of Melbourne and CEPR (email: sushant.acharya.econ@gmail.com); Benhabib: New York University and NBER (email: jess.benhabib@nyu.edu). Benjamin Moll was the coeditor for this article. Author names in this article's byline are ordered alphabetically. The authors thank Keshav Dogra, Greg Kaplan, Ricardo Reis, Gianluca Violante, Shu Lin Wee, Ricardo Lagos, and Diego Perez for comments and suggestions. We also thank seminar participants at the Richmond Fed, the Minneapolis Fed, and OzMac 2024 for helpful comments. This paper was previously circulated with a different title: "Global Indeterminacy in HANK Economies."

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features affect monetary transmission and whether the aforementioned central tenet continues to hold. The fast-growing heterogeneous agent New Keynesian (HANK) literature seeks to address this.¹ We contribute to this literature by showing that no matter how aggressively monetary policy responds to changes in inflation, HANK economies with countercyclical risk are susceptible to self-fulfilling fluctuations or “*endogenous* demand shocks.” This is true even when risk is mildly countercyclical. To understand the key force behind this inherent instability, suppose that absent any change in fundamentals, households suddenly entertain the belief that the economy will enter a recession. If risk is countercyclical, this lower level of economic activity implies that households expect to face higher risk in the future and increase their desired level of precautionary savings to self-insure against the higher probability of future declines in consumption. This higher desire to save puts downward pressure on the *natural* interest rate. If monetary policy keeps policy rates unchanged in the face of a lower natural rate, households reduce their current spending. In the presence of nominal rigidities, this lower spending translates into lower output and below-target inflation, pushing the economy into a recession and rendering the initial pessimistic belief self-fulfilling. This is analogous to an “*endogenous*” negative demand shock, which results in nonfundamental fluctuations in output and inflation.

The reason why HANK economies with countercyclical risk are susceptible to self-fulfilling fluctuations is that the natural rate of interest is *endogenous* and co-moves with output in such economies.² Standard monetary policy rules, even if they satisfy the Taylor principle, allow for self-fulfilling fluctuations in the natural rate, which act as “*endogenous* demand shocks,” and result in nonfundamental fluctuations in output and inflation. Thus, standard monetary policy rules cannot implement a unique equilibrium if risk is even mildly countercyclical. To ensure a unique equilibrium, monetary policy must act to prevent these endogenous fluctuations in the natural rate. To see this, consider the same example as above, where households believe that the economy is about to enter a recession. The resulting increase in the desired level of precautionary savings pushes down the natural interest rate. Unlike above, suppose that monetary policy instead lowers nominal rates sufficiently in response to the lower natural rate. This discourages households from reducing their current spending, and output from declining, thus preventing the initial beliefs from being confirmed in equilibrium.³

More generally, we show that monetary policy rules that do not address the endogenous fluctuations in the natural rate fail to eliminate these nonfundamental

¹ See, for example, Kaplan, Moll, and Violante (2018); McKay, Nakamura, and Steinsson (2016); Auclert, Rognlie, and Straub (2024); Acharya and Dogra (2020); Ravn and Sterk (2021); Bilbiie (2025); Gornemann, Kuester, and Nakajima (2016); and Ahn et al. (2018), among others.

² In the RANK literature, the natural rate is typically defined as the notional real interest rate that would arise in an economy where all prices were flexible. Instead, our definition of the natural rate follows Keynes (1936) and is defined as the real interest rate consistent with output remaining constant at some particular level. Importantly, while this definition of the natural rate coincides with the real interest rate in the flexible-price limit in RANK economies and HANK economies with acyclical risk, the two concepts diverge when risk is countercyclical. See Section IIC for a detailed discussion.

³ In the same fashion, if households have optimistic beliefs about the economy, they perceive that they will face lower income risk, causing them to reduce their desired level of precautionary savings. This in turn puts upward pressure on the natural interest rate. If monetary policy does not raise the policy rate sufficiently in response to this upward pressure on the natural rate, the higher spending by households leads to higher output and inflation, confirming the initial optimistic beliefs. This acts like a positive endogenous demand shock.

fluctuations driven by endogenous demand shocks. While our baseline model shows this in the context of a simple inflation-targeting monetary policy rule, this characterization continues to hold even when we allow for other standard monetary policy rules that have been shown to eliminate nonfundamental fluctuations in RANK. In particular, we show that standard rules, which adjust the policy rate in response to output fluctuations, or rules that display inertial behavior, also fail to eliminate self-fulfilling fluctuations in our HANK economy with countercyclical risk. Instead, a rule that can prevent these nonfundamental fluctuations must adjust the nominal rate at least one-for-one with these endogenous fluctuations in the natural rate. In other words, the rule must satisfy a form of the Taylor principle but for natural rates. In practical terms, this characterization implies that monetary policy must not only respond aggressively to keep the private sector's inflation expectations on target, but must also act as aggressively to keep their expectations about real activity anchored. In fact, a central bank that ignores expectations about real activity may fail at keeping even inflation expectations anchored, as doing so would leave the door open to self-fulfilling beliefs driving nonfundamental fluctuations in output and inflation.

The source of equilibrium multiplicity described above is conceptually distinct from those identified in the literature on liquidity traps. In particular, Benhabib, Schmitt-Grohé, and Uribe (2001b) show that the possibility of a binding effective lower bound (ELB) on nominal rates results in multiple equilibria and global indeterminacy in RANK.⁴ By contrast, our paper purposely abstracts from an ELB in order to highlight that countercyclical risk is a distinct force driving the global indeterminacy in HANK.

Our analysis goes beyond most existing studies of determinacy in HANK economies by considering *global*, rather than *local* determinacy. The HANK literature has shown that achieving local determinacy in HANK economies can be more demanding relative to RANK economies. Acharya and Dogra (2020); Bilbiie (2025); Auclert, Rognlie, and Straub (2023); and Ravn and Sterk (2021) find that if income risk is countercyclical, then the Taylor principle is not sufficient to ensure local determinacy. However, they also show that while the simple Taylor principle fails, a “cyclical risk augmented Taylor principle,” which demands a stronger response to inflation than in RANK, is sufficient for ensuring *local* determinacy. We show that while a stronger response may be sufficient for local determinacy, it *cannot* rule out *global* indeterminacy if risk is even mildly countercyclical. The only other paper which has studied global indeterminacy in the context of HANK models with countercyclical risk is Ravn and Sterk (2021).⁵ Our characterization of global indeterminacy in the form of two steady states is complementary to theirs. However, unlike Ravn and Sterk (2021), we provide a complete analytical characterization of the dynamics through which countercyclical risk gives rise to global indeterminacy. Moreover, we also describe what kind of policy design can eliminate global indeterminacy. Importantly, we show that designing policies that simply eliminate

⁴ See also Heathcote and Perri (2018), who show that in the presence of a binding ELB, the precautionary savings motive of low-wealth households can make the expectation of high unemployment a reality.

⁵ Ravn and Sterk (2021) study a HANK economy with search frictions and also find that global indeterminacy can emerge in their economy with countercyclical income risk. They find that a second “unemployment trap” steady state with 100 percent unemployment may emerge alongside the targeted equilibrium, implying global indeterminacy in their model.

the untargeted steady state may be unable to deliver global determinacy. In particular, when we calibrate our model to capture an empirically realistic magnitude of countercyclical risk, we show that global indeterminacy manifests not just in the form of multiple steady states, but also as a stable cycle around the targeted steady state, in which the economy can get trapped permanently. We then show that a policy design that ignores the possibility of a cycle, and focuses only on eliminating the untargeted steady state, is unable to eliminate self-fulfilling fluctuations. Thus, we argue that fully characterizing the different ways in which global indeterminacy can arise—not just through an untargeted steady state—is essential for policy design, since only policies that eliminate *all* nonfundamental equilibria can guarantee global determinacy.

Our paper also contributes to the literature that studies how fiscal policy affects equilibrium determinacy in heterogeneous agent economies. Kaplan, Nikolakoudis, and Violante (2023) study multiplicity of equilibria in a flexible-price heterogeneous agent incomplete-market economy with nominal government debt. They show that equilibrium multiplicity can emerge if the government runs persistent deficits in their heterogeneous agent economy. Similarly, Bassetto and Cui (2018) and Farmer and Zabczyk (2019) study overlapping generations economies with no nominal rigidities and show that price-level indeterminacy may emerge. Brunnermeier, Merkel, and Sannikov (2020) and Miao and Su (2024) study fiscal rules that can deliver price-level determinacy in incomplete markets economies in which agents face rate-of-return risk. In contrast, we study equilibrium determinacy in heterogeneous agent economies with nominal rigidities and show that a regime with *active* fiscal policy and *passive* monetary policy can eliminate global indeterminacy.

Our paper is also related to the older literature that studied global determinacy in RANK. Benhabib, Schmitt-Grohé, and Uribe (2001a) study the global determinacy properties of RANK economies under standard monetary policy and fiscal rules. Their findings imply that the Taylor principle delivers global determinacy unless (i) households enjoy utility from holding money, and the cross-partial derivative of the utility function $\frac{\partial^2 u(c,m)}{\partial c \partial m} < 0$, or (ii) if money is an input in the production function. Benhabib and Eusepi (2005) study a RANK economy with physical capital and show that global indeterminacy in the form of periodic cycles around the targeted steady state may emerge. Global indeterminacy in our HANK economy arises due to conceptually different reasons relative to these papers since (i) our HANK economy abstracts from the presence of money by considering the cashless limit and (ii) global indeterminacy emerges in our economy even when capital is not a factor of production. Instead, global indeterminacy arises in our HANK economy because of the presence of countercyclical risk.

Finally, Beaudry, Galizia, and Portier (2020) argue that alongside business cycle shocks, a substantial part of business cycle fluctuations can be explained by deterministic boom-bust cycles. As in our paper, they show that their New Keynesian model with financial frictions and countercyclical risk premium features deterministic limit cycles via a Hopf bifurcation, and that this provides a good description of US business cycle data. Instead, our focus is on understanding how countercyclical income risk in a HANK economy can lead to global indeterminacy.

The rest of the paper is organized as follows. Section I describes the model environment, while Section II characterizes equilibrium. Section III studies local and global determinacy under a standard inflation-targeting rule. Section IVA identifies the root of global indeterminacy and proposes a monetary policy rule that implements a unique equilibrium, and Section V concludes.

I. Model

HANK models are typically not analytically tractable because the distribution of wealth is a state variable that evolves endogenously. To make our point in the clearest possible way, we use an analytically tractable HANK model in continuous time. We achieve analytical tractability in our HANK model by assuming that utility is quasi-linear, which ensures that the aggregate dynamics of output and inflation can be characterized independently of the dynamics of the distribution of wealth. For simplicity, we abstract from aggregate risk.

A. Households

There is a continuum of households indexed by $j \in [0, 1]$. Each household has identical preferences, and the expected discounted lifetime utility of household j at date t can be written as

$$V_j(t) = \max_{\{c_{j,\tau}, n_{j,\tau}\}_{\tau=t}^{\infty}} E_t \int_t^{\infty} e^{-\rho(\tau-t)} \left\{ \frac{c_{j,\tau}^{1-\gamma^{-1}}}{1-\gamma^{-1}} - \psi n_{j,\tau} \right\} d\tau,$$

where $c_{j,\tau}$ and $n_{j,\tau}$ denote the household's date τ consumption and hours worked respectively. γ measures the elasticity of intertemporal substitution. The household's choices at all dates must satisfy the budget constraint and borrowing constraint:

$$(1) \quad \frac{da_{j,t}}{dt} = (i_t - \pi_t) a_{j,t} + w_t \xi_{j,t} n_{j,t} + D_t + T_t - c_{j,t} \quad \text{with} \quad a_{j,t} \geq -\underline{a}.$$

Households can trade a short-term risk-free nominal bond with return i_t . The real value of bond holdings of household j at time t are denoted by $a_{j,t}$. Household j supplies $\xi_{j,t} n_{j,t}$ effective labor hours and earns labor income $w_t \xi_{j,t} n_{j,t}$, where w_t denotes the real wage. Since idiosyncratic productivity $\xi_{j,t}$ is stochastic, households face labor income risk. In particular, we assume that ξ_j follows a two-state Poisson process, $\xi_j \in \{\xi_l, \xi_h\}$, where $\xi_h > \xi_l$. In addition to labor income, each household also receives dividends from firms. For simplicity, we assume that all households receive an equal share of dividends D_t and transfers from the government T_t . P_t denotes the aggregate price level in the economy. In addition, each household faces a borrowing constraint, which states that their wealth cannot fall below $-\underline{a}$, where $\underline{a} \geq 0$.

$\lambda_{l,t}$ denotes the rate at which a household with productivity ξ_h switches to productivity ξ_l , while $\lambda_{h,t}$ denotes the rate at which a household with productivity ξ_l switches to a ξ_h at date t . We allow the switching intensities to vary over time to

capture the notion that households face different levels of income risk during economic expansions and contractions. In particular, we assume that

$$(2) \quad \lambda_{l,t} = \bar{\lambda}_l y_t^{-\Theta},$$

where Θ controls the *cyclicality of income risk*. $\Theta > 0$ implies that the likelihood of ξ_h households transitioning to the low productivity state is lower when output is higher.⁶

Taking some artistic liberty and treating the ξ_l state as “unemployment,” the specification above would imply that households face a lower risk of becoming unemployed during economic expansions. In contrast, $\Theta = 0$ corresponds to *acyclical income risk*: The probability of transitioning from the ξ_h to ξ_l state is independent of the level of economic activity.⁷

Since the rate at which households transition from productivity ξ_h to ξ_l depends on y_t , the fraction of households with productivity ξ_l (given by η_t) and with productivity ξ_h (given by $1 - \eta_t$) would change with the level of economic activity if $\lambda_{h,t}$ was constant. To avoid this complication, we assume that the rate $\lambda_{h,t}$ adjusts to satisfy $\lambda_{h,t}\eta_t = \lambda_{l,t}(1 - \eta_t)$, ensuring that $\eta_t = \bar{\eta}$ for all dates t . This assumption implies that $\lambda_{h,t} = \bar{\lambda}_h y_t^{-\Theta}$, where $\bar{\lambda}_h$ is a constant and the fraction of ξ_l households at any date is given by $\bar{\eta} = \bar{\lambda}_l / (\bar{\lambda}_l + \bar{\lambda}_h)$. This assumption is made for expositional clarity; Supplemental Appendix E.5 relaxes this assumption and shows that allowing η_t to vary over time does not change our conclusion.

B. Firms

There is a continuum of monopolistically competitive firms indexed by $k \in [0, 1]$. At any date t , firm k produces a differentiated intermediate good $y_{k,t}$, which it sells to a representative final-goods firm. The final-goods firm combines the varieties using a CES aggregator to produce the final-good y_t :

$$(3) \quad y_t = \left[\int_0^1 y_{k,t}^{\frac{\varepsilon-1}{\varepsilon}} dk \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

which yields the standard demand system for each variety k :

$$(4) \quad y_{k,t} = \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon} y_t.$$

⁶Our results do not depend on the precise functional form of how this transition rate depends on the level of y_t . We discuss the robustness of our results to the particular functional form in Section IIIC.

⁷While countercyclical income risk is arguably the empirically relevant benchmark, there is no consensus on the exact measure of countercyclicality. For example, Storesletten, Telmer, and Yaron (2004) find that the standard deviation of persistent shocks to log household income increases from 0.12 to 0.21 as the aggregate economy moves from peak to trough. However, Guvenen, Ozkan, and Song (2014) find that the variance of income is acyclical, but the left skewness is countercyclical. More recently, Nakajima and Smirnyagin (2019), using a broader definition of income, find that both the variance and left skewness is countercyclical. Our formulation is consistent with both the variance and left skewness being countercyclical.

Each intermediate goods producer uses a linear production function $y_{k,t} = \ell_{k,t}$, where $\ell_{k,t}$ denotes the effective units of labor employed by firm k . Then, the period t profit of firm k can be written as $D_{k,t} = (P_{k,t}/P_t)y_{k,t} - (1 - \tau)w_t y_{k,t} - \mathcal{T}_{k,t}$, where τ denotes a payroll subsidy, which we set to $\tau = \varepsilon^{-1}$ to eliminate the average monopolistic markup. Furthermore, this payroll subsidy is financed by imposing a lump-sum tax $\mathcal{T}_{k,t} = \tau w_t y_{k,t}$, which the firm treats as given. In symmetric equilibrium, $y_{k,t} = y_t$ and dividends of any firm $k \in [0, 1]$ can be written as $D_t = (1 - w_t)y_t$. Finally, nominal rigidities are captured by a forward-looking Phillips curve:

$$(5) \quad \dot{\pi}_t = \rho\pi_t - \kappa(w_t - 1),$$

where $w_t - 1$ denotes the deviation of marginal cost from the flexible-price benchmark (in which $w_t = 1$ for all t), and $\kappa > 0$ captures the slope of the Phillips curve. While we postulate this simple linear Phillips curve (5) in our baseline model for tractability, Supplemental Appendix E.3 shows that our results continue to hold if we instead utilize a nonlinear microfounded Phillips curve as in Rotemberg (1982).

C. Monetary and Fiscal Policy

Monetary Policy.—Monetary policy sets the nominal rate i_t according to a standard interest rate rule. While we consider other standard monetary policy rules in Supplemental Appendix E.1 and E.2, in our baseline model, we specify monetary policy as a simple inflation-targeting rule:

$$(6) \quad i_t = \bar{r} + \phi_\pi \pi_t, \quad \text{where } \phi_\pi > 1.$$

$\phi_\pi > 1$ in (6) denotes how aggressively the central bank raises the nominal rate when inflation is above its steady state level of 0. As is standard in the RANK literature, we set the intercept to $\bar{r}$, which denotes the real interest rate in the flexible-price limit of our economy. As Supplemental Appendix A.3.1 shows, since we abstract from aggregate shocks, the unique equilibrium in the flexible-price limit features a constant level of output $y_t = 1$ and real interest rate $\bar{r}$ at each date t . Throughout this paper, we will require that the real interest rate in the targeted steady state is positive: $\bar{r} > 0$.

Finally, it is important to point out that (6) does not impose an ELB on nominal rates. We purposely make this choice to highlight that the source of multiplicity that we uncover is conceptually different from that in Benhabib, Schmitt-Grohé, and Uribe (2001b), where multiple equilibria arise due to the presence of an ELB.

Fiscal Policy.—For simplicity, in our baseline model we set government expenditures to zero and assume that the government runs a balanced budget at each date: $\tau w_t y_t = T_t$, where $\tau w_t y_t$ is the payroll subsidy paid out to firms, and T_t is the lump-sum tax on the same firms. We extend our analysis to include government debt in Section IVB, where we study non-Ricardian fiscal policy rules.

II. Equilibrium

A. Household Decisions

Since households have quasi-linear preferences, the date t optimal-consumption decision of household j with idiosyncratic productivity ξ_j and wealth a_j can be written as

$$(7) \quad c_t(a_j, \xi_j) = \left(\frac{\xi_j w_t}{\psi} \right)^\gamma.$$

Equation (7) shows that the optimal consumption of household j does not depend on their wealth. This insight that, with quasi-linear preferences, the consumption of a household does not depend on its wealth position has been known at least since the seminal paper by Scheinkman and Weiss (1986), and has since been used extensively in environments with heterogeneous agents and/or incomplete markets literature as it affords analytical tractability when the wealth distribution is nondegenerate (see, e.g., Lagos and Wright (2005), among others).⁸

Given our assumption of quasi-linear preferences, since the consumption of a household is independent of its wealth position, in equilibrium, all households with idiosyncratic productivity $\xi_j, j \in \{l, h\}$ enjoy the same level of consumption. We refer to the date t consumption of a household with productivity $\xi_j, j \in \{l, h\}$ as $c_{j,t}$. Normalizing $\psi = [(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma]^{1/\gamma}$, we have

$$(8) \quad c_{h,t} = \frac{\xi_h^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} w_t^\gamma,$$

and

$$c_{l,t} = \frac{\xi_l^\gamma}{(1 - \bar{\eta}) \xi_h^\gamma + \bar{\eta} \xi_l^\gamma} w_t^\gamma,$$

where $c_{h,t} > c_{l,t}$. While the consumption of each household does not depend on their wealth, the amount of leisure they enjoy does depend on how wealthy they are. Comparing two households with the same idiosyncratic productivity, the household with higher wealth tends to enjoy more leisure. Furthermore, Supplemental Appendix A.1 shows that for a given real interest rate, the expected consumption growth of households with productivity ξ_l is always greater than that of households with productivity ξ_h . Thus, in equilibrium, all households with productivity ξ_l are

⁸See also Bilbiie (2008), who does not use quasi-linear preferences in his model, but, in footnote 14, discusses how assuming quasi-linear preferences makes household heterogeneity redundant in his model, as these preferences imply that the consumption decisions of savers in his model do not depend on wealth. Similarly, while Bilbiie (2025) does not impose quasi-linear preferences throughout the paper, footnote 11 of Bilbiie (2025) also discusses the implications of imposing quasi-linear preferences in that paper's model: With quasi-linear preferences, inequality is acyclical in that model because these preferences imply that the consumption decisions of savers in his model do not depend on wealth.

borrowing constrained. In contrast, ξ_h households are *on* their Euler equation, which can be written as

$$(9) \quad \frac{\dot{c}_{h,t}}{c_{h,t}} = \underbrace{\gamma(i_t - \pi_t - \rho)}_{\text{intertemporal-substitution}} + \underbrace{\gamma\lambda_{l,t} \left[\left(\frac{c_{l,t}}{c_{h,t}} \right)^{-\frac{1}{\gamma}} - 1 \right]}_{\text{precautionary savings}}.$$

The first term on the right-hand side of (9) shows that the consumption growth of unconstrained households depends positively on the real interest rate $r_t = i_t - \pi_t$: the *intertemporal-substitution* channel. A higher real interest rate, holding all else constant, incentivizes the unconstrained households to delay consumption, raising their consumption growth. The second term on the right-hand side of (9) captures the *precautionary-savings* motive. The larger the drop in consumption when a ξ_h household transitions to the ξ_l state (smaller c_l/c_h), the stronger is this motive. Similarly, holding c_l/c_h fixed, a higher risk aversion (smaller γ) or a higher probability of transitioning from ξ_h to ξ_l (higher $\lambda_{l,t}$) also strengthens this motive and increases consumption growth for any given real interest rate.

B. Market Clearing

Goods market clearing requires that the amount of final good produced must be consumed:

$$(10) \quad y_t = (1 - \bar{\eta})c_{h,t} + \bar{\eta}c_{l,t},$$

where the right-hand side of (10) is equal to aggregate consumption. Using (8), Supplemental Appendix A.2 shows that (10) implies a log-linear relationship between aggregate output and wages:

$$(11) \quad w_t = y_t^{\frac{1}{\gamma}}.$$

Since wages $w = 1$ in the targeted steady state, (11) also implies that the steady state level of output in the targeted steady state is $y = 1$. Using (8) and (11) also implies that $c_{h,t}$ and $c_{l,t}$ can be rewritten as

$$(12) \quad c_{j,t} = \frac{\xi_j^\gamma}{(1 - \bar{\eta})\xi_h^\gamma + \bar{\eta}\xi_l^\gamma} y_t, \quad \text{where } j \in \{h, l\} \Rightarrow \frac{c_{h,t}}{c_{l,t}} = \left(\frac{\xi_h}{\xi_l} \right)^\gamma.$$

While consumption of both ξ_h and ξ_l households co-moves with output, the ratio $c_{h,t}/c_{l,t} > 1$ is constant over time. Finally, at any date t , since all ξ_l households are borrowing constrained and have $a = -\underline{a}$, asset market clearing requires that asset holdings of ξ_h households as a whole are given by $\bar{\eta}\underline{a}$.

C. The Neutral Rate and the Natural Rate of Interest

As is standard in the textbook treatment of the RANK model, the aggregate dynamics of output and inflation in our tractable HANK economy can be also summarized

by an IS curve (investment-saving) curve and a Phillips curve. Using (11) in (5), we can express the Phillips curve in terms of output and inflation:

$$(13) \quad \dot{\pi}_t = \rho\pi_t - \kappa\left(y_t^{\frac{1}{\gamma}} - 1\right).$$

Supplemental Appendix A.3.1 shows that using (12) and (2) in (9), the “IS curve” in our HANK economy is⁹

$$(14) \quad \frac{\dot{y}_t}{y_t} = \gamma(i_t - \pi_t - r^*(y_t)), \quad \text{where} \quad r^*(y) = \rho - \sigma y^{-\Theta}.$$

As in the textbook three-equation RANK model, (14) shows that output growth is positive when monetary policy implements a real interest rate $r_t = i_t - \pi_t$, which is higher than *natural rate of interest* $r^*(y_t)$. Here, we define $r^*(y)$ as the real interest rate, which sets $\dot{y}_t/y_t = 0$ and $y_t = y$ for all t —that is, $r^*(y)$ is the real interest rate, which is consistent with output remaining constant at $y_t = y$ for all t .

In the RANK limit of our model ($\bar{\lambda}_l = 0$), since households do not face consumption risk, and because we abstract from aggregate shocks, the natural rate is constant over time and simply equal to the discount rate: $r^*(y) = \rho$. This means that setting $r = r^* = \rho$ in RANK is consistent with output remaining fixed at any level of output y . The *natural rate* $r^*(y)$ also coincides with the real interest rate in the flexible-price limit of our economy, $\bar{r} = r^*(y) = \rho$. The same is true in HANK if risk is acyclical ($\Theta = 0$). The only difference is that now both the natural rate and the flexible-price real interest rate are lower than in RANK and are given by $r^*(y) = \bar{r} = \rho - \sigma$, where $\sigma = \bar{\lambda}_l(\xi_h/\xi_l - 1)$ measures the expected increase in marginal utility when a ξ_h household transitions to lower idiosyncratic productivity ξ_l and captures the fact that households face consumption risk. Clearly, $\sigma = 0$ if the probability of transitioning to the low productivity state is zero ($\bar{\lambda}_l = 0$), or if the consumption across the two idiosyncratic productivity levels is the same (which occurs if $\xi_h = \xi_l$). In all other cases, $\sigma > 0$. Since households face consumption risk, they save for precautionary reasons, and this requires a lower interest rate for the asset market to clear.

However, if risk is countercyclical ($\Theta > 0$), the natural rate is *endogenous* and co-moves with output:

$$(15) \quad \frac{dr^*(y)}{dy} = \sigma\Theta y^{-(1+\Theta)} > 0;$$

⁹In equilibrium, while the actual value of the debt limit $\underline{a}$ does not affect aggregate dynamics of y_t and π_t , it does matter for the wealth distribution that emerges in equilibrium. However, quasi-linear preferences render our economy block recursive, allowing us to characterize the dynamics of output and inflation independently of the wealth distribution.

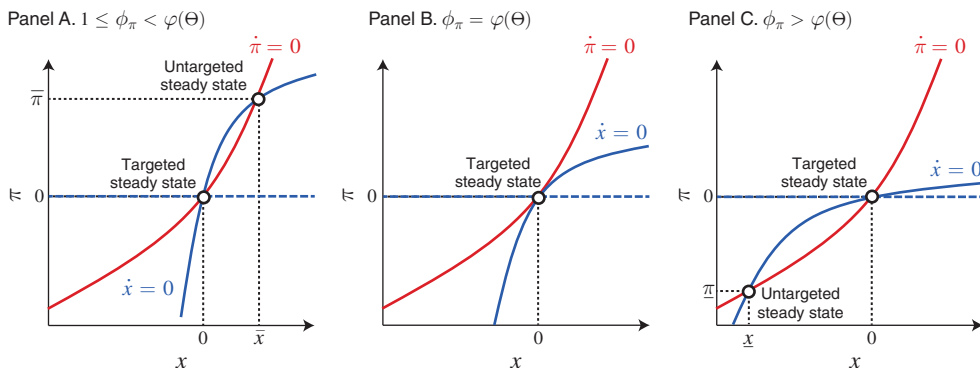
that is, the real interest rate consistent with output being constant at a particular level y , now depends on the level of output itself. Moreover, the natural rate $r^*(y)$ is an increasing function of y . Thus, in our HANK economy with countercyclical risk, the flexible-price real interest rate $\bar{r} = \rho - \sigma$ is, in general, different than the natural rate $r^*(y)$. The two concepts only coincide in steady state when $y = 1$: $\bar{r} = r^*(1)$. To see why, recall that if output was lower than $y = 1$, countercyclical risk implies that ξ_h households face greater consumption risk, as now they face a larger chance of switching to the ξ_l state. This causes them to reduce their current consumption demand and increase their precautionary savings. This greater desire to save implies that a lower real interest rate is required to clear asset markets and keep demand constant at that lower level.

Notice that our definition of the natural rate of interest $r^*(y)$ differs from how the term “natural rate” is used in the context of the textbook three-equation New Keynesian model (see, e.g., Woodford 2003; Galí 2015). In the New Keynesian literature, the natural rate is typically defined as the real interest rate that would prevail in the flexible-price limit of the economy, which is equal to $\bar{r} = r^*(1) = \rho - \sigma$ in our HANK model. This flexible-price real interest rate depends on exogenous parameters and potentially varies over time only in response to *exogenous* shocks—for example, shocks to the discount rate ρ . Importantly, it does not depend on *endogenous* variables such as the level of output. Consequently, in RANK and in HANK with acyclical risk, our definition of “natural rate” coincides with the flexible-price real interest rate. In contrast, when risk is countercyclical $\Theta > 0$, the natural rate $r^*(y)$, as we define it, *does* depend on *endogenous* variables, specifically output: In a weak economy, where output is below its flexible-price level, a lower real interest rate is required to maintain demand, and hence output, at that level. In contrast, even in our economy with $\Theta > 0$, the flexible-price real interest rate does not depend on endogenous variables, $\bar{r} = \rho - \sigma$. Thus, in our HANK economy with countercyclical risk, our definition of the natural rate does not always coincide with the flexible-price real interest rate: There are many *natural* rates $r^*(y)$, one for a given level of y , but there is a unique flexible-price real interest rate $\bar{r}$, which coincides with the natural rate consistent with $y = 1$. Our choice of terminology hearkens back to Keynes (1936, 242–243):

For every rate of interest there is a level of employment for which that rate is the “**natural**” rate, in the sense that the system will be in equilibrium with that rate of interest and that level of employment. ... we might term the **neutral** rate of interest, ... the natural rate in the above sense which is consistent with full employment, given the other parameters of the system.
[emphasis ours]

D. Aggregate Dynamics and Steady States

Given the monetary policy rule (6), Proposition 1 below shows that the aggregate dynamics in our HANK economy are described by the two-dimensional system of ordinary differential equations. However, rather than describing the dynamics of output, it is more convenient to characterize the dynamics of the (scaled) *output gap*, which we define as the log deviation of output from its flexible-price level scaled by γ^{-1} : $x = \gamma^{-1}(\ln y - \ln 1)$.

FIGURE 1. MULTIPLE STEADY STATES WITH $\Theta > 0$

PROPOSITION 1: *Given the interest rate rule (6), the aggregate dynamics of x_t, π_t can be written as*

$$(16a) \quad \dot{x}_t = (\phi_\pi - 1) \pi_t - (r^*(x_t) - \bar{r}),$$

$$(16b) \quad \dot{\pi}_t = \rho \pi_t - \kappa(e^{x_t} - 1),$$

where $r^*(x_t) - \bar{r} = \sigma(1 - e^{-\gamma\Theta x_t})$ denotes the difference between the natural rate of interest and neutral rate of interest. The target steady state is given by $x = \pi = 0$.

PROOF:

See Supplemental Appendix A.3.1.

Equations (16a) and (16b) nest the RANK benchmark: In RANK, $\sigma = 0$, and so $r^*(x_t) = \bar{r} = \rho$ for any x , implying that the last two terms on the right-hand side of (16a) eliminate each other. The same is true in HANK if risk is acyclical ($\Theta = 0$), even though the natural and neutral rates of interest are both lower in this case than in RANK. Thus, (16a) and (16b) show that the global dynamics in the RANK benchmark and in a HANK economy with acyclical ($\Theta = 0$) are identical as long as ϕ_π is the same in both economies. Thus, as has also been pointed out by Werning (2015), simply the presence of risk does not necessarily alter the dynamics of output and inflation. In contrast, when risk is countercyclical, (16a) reveals that an extra force shapes global dynamics relative to the RANK and acyclical risk benchmark. In particular, fluctuations in the output gap drive an *endogenous* gap between the natural rate and neutral rate of interest, which in turn feeds back into the dynamics of output and inflation.

The three panels of Figure 1 plot the nullclines associated with (16a) and (16b) in (x, π) space for different values of ϕ_π . The solid red line depicts the $\dot{\pi}_t = 0$ nullcline, which is unaffected by the presence of risk or the cyclicity of risk. In contrast, in the RANK benchmark $\sigma = 0$ (or in HANK with $\Theta = 0$), the $\dot{x} = 0$ nullcline (equivalently, the long-run IS) is depicted by the dashed horizontal curve at $\pi = 0$,

while in HANK with $\Theta > 0$, it is depicted by the upward-sloping solid blue curve. The reason that the nullcline is flat in RANK (or in HANK with $\Theta = 0$) is that $r^*(x)$ is constant, but when risk is countercyclical, $r^*(x)$ co-moves with output and is thus upward sloping.

Any intersection of the nullclines constitutes a steady state. Clearly, both nullclines intersect at $x = \pi = 0$, implying that the *targeted steady state* always exists, in which output equals its flexible-price level ($x = 0$) and inflation is on target ($\pi = 0$). In fact, as Figure 1 shows, this is the only steady state in the RANK limit or when risk is acyclical. In this case, the long-run IS curve is depicted by the horizontal dashed blue curve and only intersects the $\dot{\pi} = 0$ nullcline at the targeted steady state. However, when risk is countercyclical ($\Theta > 0$), the long-run IS curve intersects the $\dot{\pi} = 0$ nullcline twice, implying that an additional *untargeted steady state* emerges (see Supplemental Appendix B.3 for a proof). As we discuss next, whether output is above or below its level in the targeted steady state in this untargeted steady state depends on how aggressive monetary policy is (magnitude of ϕ_π).

The emergence of the second steady state is rooted in the fact that, with countercyclical risk, households face greater risk when output is lower. Let's start by considering the case with $1 < \phi_\pi < \varphi(\Theta)$ (depicted graphically in Figure 1, panel A). In this case, the untargeted steady state features a positive output gap $\bar{x} > 0$, and above-target inflation $\bar{\pi} > 0$. To see why such a steady state emerges, suppose that households believe that the economy will have higher output, $\bar{x} > 0$, forever. Because risk is countercyclical, this belief about higher output also implies that households perceive that they face lower income risk. This causes them to reduce their precautionary savings demand and to increase consumption spending. Owing to the presence of nominal rigidities, this higher spending puts upward pressure on output and inflation. When $1 < \phi_\pi < \varphi(\Theta)$, the monetary policy rule (6) raises nominal rates in response to the higher inflation, but the implied increase in real interest rates is not sufficient to dissipate this higher demand, thus allowing the beliefs of higher output to become self-fulfilling.

Raising ϕ_π toward $\varphi(\Theta)$ induces a larger increase in nominal and hence real interest rates. The higher real rate lowers output in the second steady state, bringing $\bar{x}$ closer to 0. Graphically, a higher ϕ_π shifts the solid blue long-run IS curve lower and makes the untargeted steady state shift lower down on the $\dot{\pi} = 0$ nullcline, closer to $(0, 0)$. In fact, as one increases ϕ_π all the way to $\varphi(\Theta)$, the long-run IS curve becomes tangent to the $\dot{\pi} = 0$ nullcline at the targeted steady state. This is depicted graphically in Figure 1, panel B. In this knife-edge case, the only steady state is the targeted steady state $x = \pi = 0$.

Increasing ϕ_π even further ($\phi_\pi > \varphi(\Theta)$) shifts the long-run IS curve down further (depicted in Figure 1, panel C), and multiple steady states emerge again. The untargeted steady state now features lower output $\bar{x} < 0$ and below-target inflation $\bar{\pi} < 0$. With countercyclical risk, lower output ($\bar{x} < 0$) implies that households face more risk in this steady state compared to the targeted steady state, prompting them to increase their precautionary savings. This causes them to lower spending, which puts downward pressure on output and inflation. Monetary policy, following the rule (6), lowers the nominal rate in response to the lower inflation, but despite this, the equilibrium real interest rate remains too high to discourage the higher precautionary savings. This reinforces lower household demand, trapping the economy

at a lower level of economic activity $\bar{x} < 0$. Supplemental Appendix B.3 shows that this untargeted steady state persists even if we keep raising ϕ_π further (as long as it remains finite). Increasing ϕ_π further only has the effect of making output in the untargeted steady state even lower.

While the discussion above shows that in our HANK economy with countercyclical risk, the untargeted steady state can feature output which is higher, lower, or equal to that in the targeted steady state; in what follows, we will focus on the scenario in which the targeted steady state is *locally determinate*. As we show in Section III, this requires that $\phi_\pi > \varphi(\Theta)$, implying that the untargeted steady state always exists and features lower output and inflation than in the targeted steady state (Figure 1, panel C).

III. Local versus Global Determinacy of Equilibrium

In order to distinguish between local and global determinacy, it is helpful to rewrite (16a)–(16b) as

$$(17) \quad \begin{bmatrix} \dot{x}_t \\ \dot{\pi}_t \end{bmatrix} = \underbrace{A \begin{bmatrix} x_t \\ \pi_t \end{bmatrix}}_{\text{first-order terms}} + \underbrace{\begin{bmatrix} \sigma \sum_{s=2}^{\infty} (-1)^s \frac{\gamma^s \Theta^s}{s!} x^s \\ -\kappa \sum_{s=2}^{\infty} \frac{1}{s!} x^s \end{bmatrix}}_{\text{higher-order terms}}, \quad A = \begin{bmatrix} -\sigma\gamma\Theta & \phi_\pi - 1 \\ -\kappa & \rho \end{bmatrix}.$$

Rewriting (16a)–(16b) in this way highlights that the dynamics of (x_t, π_t) local to the targeted steady state $(0,0)$ are dominated by the first-order terms, while the higher-order terms dominate the dynamics when the economy is further away from the targeted steady state.¹⁰

Local determinacy requires that any trajectory, other than $(x_t, \pi_t) = (0,0)$, that starts in a small neighborhood of the targeted steady state $(0,0)$ leaves this neighborhood—that is, all these trajectories *do not remain bounded* inside this neighborhood. The only trajectory that stays bounded in this neighborhood is $(x_t, \pi_t) = (0,0)$, implying that $(0,0)$ is the unique *bounded* equilibrium in its neighborhood. In other words, if one limits analysis to rational expectations equilibria in which (x, π) remain forever in a small neighborhood of the targeted steady state $(0,0)$, then the only bounded equilibrium is $(0,0)$. Since the first-order terms dominate the behavior of x_t, π_t in this small neighborhood around the targeted steady state, whether the economy is locally determinate around the targeted steady state depends on the eigenvalues of the matrix A . In particular, we need both eigenvalues of A to be *explosive*—that is, they have positive real parts. Proposition 2 formally describes when this is the case.

¹⁰It is worth noting that up to first order, the IS equation in (17) is very similar to those derived in (or would be implied by) recent papers that present analytically tractable HANK models—for example, Werning (2015); Challe (2020); Acharya and Dogra (2020); Ravn and Sterk (2021); Bilbiie (2025), among others.

PROPOSITION 2 (Local Determinacy in HANK with Countercyclical Risk): *The targeted equilibrium of the economy described by (16a)–(16b) is locally determinate if ϕ_π satisfies*

$$(18) \quad \phi_\pi > \varphi(\Theta), \quad \text{where} \quad \varphi(\Theta) = 1 + \frac{\rho\sigma\gamma\Theta}{\kappa},$$

provided that risk is not too countercyclical $\Theta \in [0, \Theta^)$, where $\Theta^* \equiv \rho/\sigma\gamma$. If $\Theta > \Theta^*$, then the targeted equilibrium is locally indeterminate for any finite ϕ_π , no matter how large it is.*

PROOF:

See Supplemental Appendix B.2.

Equation (18) is the analog of the “cyclical risk” augmented Taylor principle derived in Acharya and Dogra (2020); Auclert, Rognlie, and Straub (2023); and Bilbiie (2025), in the context of our model, and states that as long as risk is not too countercyclical, a large enough ϕ_π that satisfies (18) ensures local determinacy. This condition simplifies to the standard Taylor principle $\phi_\pi > 1$ in the RANK limit of our model ($\sigma = 0$) and also when households face risk ($\sigma > 0$), but this risk is acyclical ($\Theta = 0$). However, when risk is countercyclical $\Theta > 0$, (18) shows that monetary policy needs to respond more aggressively to changes in inflation, the more countercyclical risk is.

To see why a larger ϕ_π is needed to ensure local determinacy with countercyclical risk, it is useful to first understand why $\phi_\pi > 1$ ensures local determinacy in RANK. As Cochrane (2011) explains: Absent the zero lower bound, the Taylor principle ensures a unique bounded equilibrium in RANK, because off equilibrium, “higher inflation leads the Fed to set interest rates in a way that produces even higher future inflation.” In other words, imposing the Taylor principle induces explosive dynamics if the economy is not on the targeted equilibrium, thus leaving the targeted equilibrium as the *unique bounded equilibrium* in RANK. To see this in our model, imposing the RANK limit $\sigma = 0$ in (16a), the IS curve is given by

$$\dot{x}_t = (\phi_\pi - 1) \pi_t.$$

With $\phi_\pi > 1$, the expression above shows that $\dot{x}_t$ is *increasing* in π_t —that is, any deviation of inflation from its target induces destabilizing dynamics, causing x to change. For example, consider a case in which inflation is below target at date 0, $\pi < 0$. Then, if $\phi_\pi > 1$, the IS curve (16a) implies that x_t must decline. Since the Phillips curve (16b) implies that inflation at any date is the net present discounted value of future marginal costs, π falls further below target over time.¹¹ These

¹¹ Output declines following date 0 unambiguously, but whether it declines monotonically over time or not depends on the eigenvalues of the system. While the Taylor principle ensures that the two eigenvalues are positive, the eigenvalues can either both be real or both complex with positive real parts. When the eigenvalues are real, x continues to decline monotonically towards $-\infty$, while if the two roots are complex, the trajectory of x is oscillatory with ever increasing amplitude. However, in both cases, as long as the Taylor principle is satisfied, the trajectory for x diverges away from its targeted level of 0 over time. Whether inflation diverges from its targeted level monotonically or in an oscillatory fashion depends on whether output declines monotonically or diverges in an oscillatory fashion, which in turn depends on the eigenvalues of the system.

destabilizing dynamics induced by the Taylor principle ensure that any trajectory that originates at any point other than $(0,0)$ does not remain bounded, and hence is not a valid equilibrium. In fact, the larger is ϕ_π relative to 1, the more destabilizing these dynamics are. Consequently, $(x, \pi) = (0,0)$ is the only *locally bounded* equilibrium in RANK if the Taylor principle is satisfied.

In contrast, when risk is countercyclical, the IS equation (16a) can be written as

$$(19) \quad \dot{x} = \underbrace{(\phi_\pi - 1)\pi - \sigma\gamma\Theta x}_{\text{first-order terms}} + \text{higher-order terms},$$

and since local determinacy depends on the behavior of (x, π) local to the targeted steady state $(0,0)$, only the first-order terms of the IS curve matter for local determinacy. When $\phi_\pi > 1$, the first linear term on the right-hand side of (19) still induces destabilizing dynamics as in RANK. However, in HANK with countercyclical risk, $\Theta > 0$, the natural rate $r^*(x)$ endogenously *co-moves* with x . This is reflected in the extra first-order term $-\sigma\gamma\Theta x$ on the right-hand side, which induces *stabilizing* dynamics because the coefficient on x is negative: A positive x causes $\dot{x}$ to become negative—that is, it causes x to return to 0, given all else. Thus, when risk is countercyclical, ϕ_π needs to be larger so that the destabilizing effect overwhelms this stabilizing effect. The cyclical risk augmented Taylor principle (18) states how large ϕ_π needs to be for this to be the case. Since this stabilizing force is not present in RANK or in HANK with acyclical risk, $\phi_\pi > 1$ suffices.

Global determinacy is more demanding than local determinacy as it requires that any trajectory, starting at any $(x, \pi) \in (-\infty, \infty)^2$ other than $(x, \pi) = (0,0)$, *does not remain bounded*. Thus, global determinacy does not just depend on the behavior of x, π local to the targeted steady state. Consequently, global determinacy depends not just on the first-order terms but also the higher-order terms in (17). In linear models, local determinacy implies global determinacy as there are no higher-order terms when the model is actually linear. However, in nonlinear models, local determinacy *need not* imply global determinacy. This is because local determinacy only ensures that the trajectories starting in a small neighborhood of the targeted equilibrium diverge away from this equilibrium. It, however, does not ensure that once this trajectory takes the economy further away from the targeted equilibrium, that the higher-order terms prevent this trajectory from growing unbounded. Thus, when one concludes that the equilibrium is unique in a nonlinear model by checking for local determinacy, one is implicitly making an assumption that a trajectory that initially diverges from the targeted equilibrium eventually becomes unbounded. However, this assumption may not hold.

Even though our model economy is nonlinear in RANK limit ($\sigma = 0$) or in the acyclical risk case ($\sigma > 0, \Theta = 0$), Supplemental Appendix B.1 shows that the standard Taylor principle $\phi_\pi > 1$, which ensures that the targeted equilibrium is locally determinate, also ensures that it is globally determinate.¹² This is because

¹²Of course, global determinacy would not obtain if we enforced an ELB since Benhabib, Schmitt-Grohé, and Uribe (2001b) show that imposing an ELB introduces multiple bounded trajectories that are consistent with equilibrium, implying that there can be global indeterminacy even when the target equilibrium is locally determinate. As aforementioned, we purposely do not impose an ELB to highlight that multiplicity of equilibria can emerge in HANK models even absent the ELB.

in the RANK limit or in HANK with acyclical risk, the IS curve does not have any higher-order terms. When risk is countercyclical, $\phi_\pi > \varphi(\Theta)$ still ensures that local to $(0,0)$, the economy features explosive dynamics.¹³ Thus, while $\phi_\pi > \varphi(\Theta)$ ensures that the first-order terms in (19) induce explosive dynamics, it cannot guarantee that the higher-order terms also do so. In fact, as we discuss next, in our HANK economy with countercyclical risk, these higher-order terms induce a stabilizing force that cannot be overwhelmed no matter how large ϕ_π is. Consequently, there exist multiple trajectories $\{x_t, \pi_t\}_{t \geq 0}$ that satisfy all equilibrium conditions and remain bounded—that is, multiple equilibria exist, implying that there is global indeterminacy. This is formalized in Proposition 3 below.

PROPOSITION 3 (Global Indeterminacy with Countercyclical Risk): *Consider the economy described in Proposition 1 for any $\Theta > 0$, and assume that $\phi_\pi > \varphi(\Theta)$. Then for any $\Theta > 0$, the equilibrium is globally indeterminate, no matter how large ϕ_π is (as long as it is finite).*

PROOF:

See Supplemental Appendix B.5.

The existence of multiple equilibria opens the door to self-fulfilling beliefs that can drive nonfundamental fluctuations and can even lead output and inflation to permanently deviate from the targeted steady state. The easiest way to see that multiple equilibria exist is to recall from Section IID that when risk is countercyclical, an untargeted steady state (which features a lower level of economic activity), exists alongside the targeted steady state. Thus, there are at least two trajectories, $\{x_t, \pi_t\}_{t \geq 0} = (0,0)$ and $\{x_t, \pi_t\}_{t \geq 0} = (\underline{x}, \underline{\pi})$, that satisfy all equilibrium conditions and remain bounded, implying that we have global indeterminacy. Even absent the ELB, households' pessimistic self-fulfilling beliefs can cause the HANK economy with countercyclical risk to *jump* from the targeted to the untargeted steady state. While this is reminiscent of the secular stagnation literature,¹⁴ which showed that the economy can get stuck at low levels of economic activity because monetary policy is stuck at the ELB, our HANK economy stagnates not because of a binding ELB, but because monetary policy fails to fully account for the endogenously lower natural rate.

However, the hazards induced by the inflation-targeting rule (6) in our HANK economy are not limited to such permanent slumps. Supplemental Appendix B.5 shows that the inability of monetary policy to fully account for endogenous fluctuations in the natural rate can also lead to other, less abrupt, nonfundamental fluctuations, the precise form of which depends on how countercyclical risk is. In particular, the global dynamics of our economy can be divided into three regimes, which we label (i) mildly countercyclical, (ii) moderately countercyclical, and (iii) highly countercyclical. Formally, we deem risk to be highly countercyclical if $\Theta > \Theta^*$, where $\Theta^* = \rho/\sigma\gamma$ is the same as in Proposition 2. Risk is said to be

¹³This is true if risk is not too countercyclical: $\Theta \in (0, \Theta^*)$. Proposition 2 already shows that when $\Theta > \Theta^*$, no matter how large ϕ_π is, monetary policy cannot induce destabilizing dynamics if the economy strays from the targeted equilibrium.

¹⁴See, for example, Benigno and Fornaro (2018); Eggertsson, Mehrotra, and Robbins (2019).

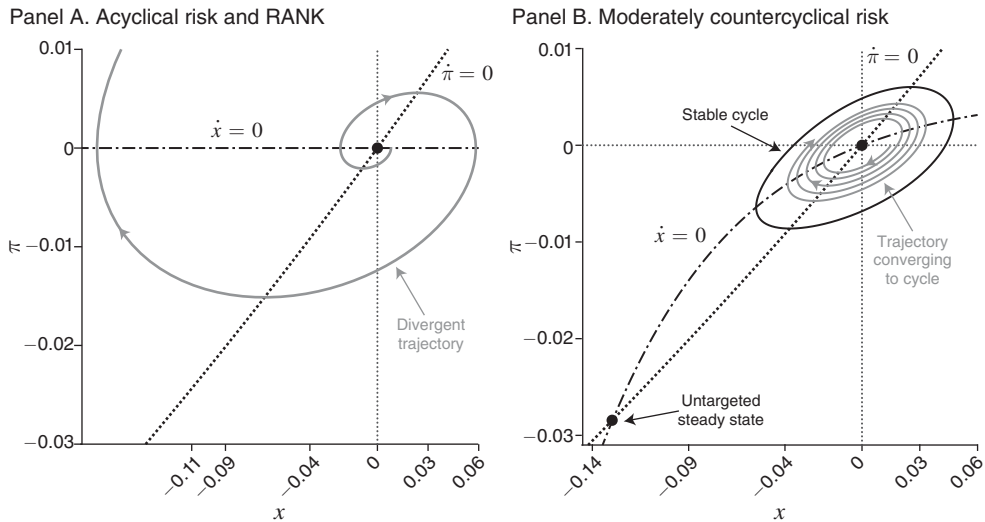


FIGURE 2. PHASE PORTRAITS WITH ACYCLICAL RISK AND COUNTERCYCLICAL RISK

Notes: In both panels, the dotted curve depicts the $\dot{\pi} = 0$ nullcline, while the dash-dotted curve depicts the $\dot{x} = 0$ nullcline. In panel A, the solid gray trajectory depicts a representative trajectory that originates near $(0,0)$ and then grows unbounded. In panel B, the solid gray trajectory depicts a representative trajectory that originates near $(0,0)$ and then converges to the stable cycle, which is depicted by the solid black trajectory. Both panels utilize the calibration described in Section IIIA.

moderately countercyclical if $\Theta \in (\Theta^\diamond, \Theta^*)$, where the threshold $\Theta^\diamond$ lies between 0 and Θ^* , and is described in Supplemental Appendix B.5. Finally, we call risk mildly countercyclical when $\Theta \in (0, \Theta^\diamond)$. Proposition 2 already shows that if risk is highly countercyclical, even with $\phi_\pi > \varphi(\Theta)$, the targeted equilibrium is locally *indeterminate*, and thus also globally indeterminate. Thus, in our discussion below, we only focus on the cases in which risk is mildly and moderately countercyclical.

When risk is mildly or moderately countercyclical, (18) ensures that the targeted steady state equilibrium is locally determinate. Consequently, any trajectory that starts in the neighborhood of $(0,0)$ initially diverges away from the targeted steady state. However, as Supplemental Appendix B.5 shows, not all of these initially divergent trajectories grow unbounded eventually. In particular, when risk is mildly countercyclical $\Theta \in (0, \Theta^\diamond)$, the stabilizing influence of the higher-order terms ensures that there exists a *saddle connection* along which the economy can transition from the neighborhood of the targeted steady state to the untargeted steady state and thus remain bounded. Furthermore, any trajectory that originates on this saddle connection also remains bounded. Similarly, when risk is moderately countercyclical, (18) still ensures that the first-order terms cause any trajectory that starts in the neighborhood of $(0,0)$ to initially diverge away from the targeted steady state (depicted by the solid gray trajectory in Figure 2, panel B). However, Supplemental Appendix B.5 shows that *none* of these initially divergent trajectories grow unbounded as the higher-order terms *push back* toward the targeted steady state. Consequently, these trajectories converge to a stable-limit cycle (depicted by the solid black trajectory in Figure 2, panel B). This shows that even a small shock

that dislodges the economy from the targeted steady state can cause the economy to get stuck in a cycle in which the output gap and inflation are permanently away from their targeted values. In contrast, in the RANK limit of our model ($\sigma = 0$), or if risk is acyclical ($\Theta = 0$), the absence of nonlinear terms in the IS equation implies that even with the standard inflation-targeting rule (6), the targeted equilibrium $(x_t, \pi_t) = (0, 0)$ for all t is the only bounded trajectory that satisfies all equilibrium conditions. This is depicted in Figure 2, panel A, which plots a trajectory starting away from $(0, 0)$, and shows that it diverges away from the targeted equilibrium and eventually becomes unbounded. Figure 2 only depicts the dynamics in the acyclical risk and moderately countercyclical cases, but Figure B.1 in Supplemental Appendix B.5 describes global dynamics for all values of Θ .

It is useful to point out that because our baseline model does not feature any predetermined variables, simply the existence of the untargeted steady state (even without characterizing these additional nonfundamental dynamics) is sufficient to establish global indeterminacy. This is because, absent any predetermined variable, the economy can just *jump* from one steady state to another. However, our characterization of the complete set of possible nonfundamental fluctuations is more than a theoretical curiosity. This is because, in Section IV, when we discuss policy design to ensure global determinacy, it will be important to design a rule that can eliminate *all* nonfundamental fluctuations, and not just the untargeted steady state. In particular, we show that simple policies that are designed to only eliminate the untargeted steady state may fail to eliminate the stable cycle that surrounds the targeted steady state. Consequently, even absent an untargeted steady state, the equilibrium is globally indeterminate because any trajectory starting in the neighborhood of the targeted steady state initially diverges but then converges to the stable cycle, hence remaining bounded.

Moreover, our characterization above helps demonstrate that our findings in the baseline model are robust to when we extend the model to include predetermined variables. Since our baseline model has no predetermined variables, the economy can stochastically jump not just between the targeted steady state and the untargeted steady state, but also between the cyclical trajectories described above in response to *sunspot* shocks. However, such jumps are not possible if the model includes a predetermined variable. To show that our characterization of global indeterminacy is robust to the inclusion of predetermined variables, in Supplemental Appendix E.5, we relax the assumption that the fraction of ξ_l households remains constant. Despite this change, an untargeted steady state with lower economic activity as well as a higher fraction of ξ_l households exists in this extension as long as risk is countercyclical. However, simply the existence of an untargeted steady state no longer implies global indeterminacy since the fraction of ξ_l households at any date is predetermined, and thus cannot *jump*. Despite this caveat, Supplemental Appendix E.5 shows that we still have global indeterminacy. In fact, the additional bounded trajectories resemble those described in our baseline model. For example, when risk is mildly countercyclical, there exists a *saddle connection* along which the economy can transition from near the targeted steady state to the untargeted steady state. The economy can no longer instantaneously jump from one steady state to another: The transition features a gradual increase in the fraction of ξ_l households alongside a gradual decline in output and inflation. Similarly, when risk is moderately

countercyclical, there exists a stable cycle that surrounds the targeted steady state. Supplemental Appendix E.5 shows that any trajectory starting in the neighborhood of the targeted steady state eventually converges to the stable cycle and remains bounded. Unlike in our baseline model, where the economy could instantaneously jump onto the stable cycle, this transition to the stable cycle is also gradual since the fraction of ξ_l households cannot jump.¹⁵

A. Quantitative Perspective on Global Indeterminacy

While Proposition 3 and the subsequent exposition make clear that the HANK economy with countercyclical risk features global indeterminacy, it is still useful to focus our discussion around the empirically relevant range of Θ . To do so, we calibrate our model to highlight what form global indeterminacy can take once we impose some discipline on the parameters.

Calibration.—In our preferred calibration, we set the discount rate ρ to be consistent with a real interest rate of 4 percent in steady state. We set the coefficient of relative risk aversion $\gamma^{-1} = 2$. We set the rate at which ξ_h households transition to ξ_l productivity $\bar{\lambda}_l = 0.013$ based on the estimates of Bilbiie, Primiceri, and Tambalotti (2023).¹⁶ Translating their estimates into continuous time yields a range for Θ between 21.98 and 29.9, with the modal estimate of 28.1. To calibrate the relative differences in idiosyncratic productivity ξ_h/ξ_l , we use estimates of the decline in consumption when a household transitions from employment to unemployment. In particular, we set $c_h/c_l = 1.1$, which is consistent with the empirical estimates of the decline in consumption when a household experiences involuntary unemployment.¹⁷ Equation (12) then implies that $\xi_h/\xi_l = 1.23$. Finally, as is common in the literature, we set $\phi_\pi = 1.5$.

Given this calibration, we can now compute the boundaries of the mildly, moderately, and highly countercyclical regions. While it is not possible to analytically characterize the value of $\Theta^\diamond$, our calibrated model features $\Theta^\diamond \approx 15.8$,

¹⁵We also introduce predetermined variables to our baseline model in two other ways. First, when we study a backward-looking rule in Supplemental Appendix E.2, a weighted average of inflation in the past acts as a predetermined variable. Since this lagged inflation measure is predetermined and cannot jump, the existence of the untargeted steady state per se does not imply global indeterminacy. However, Supplemental Appendix E.2 shows that we still have global indeterminacy, and depending on how backward looking the rule is, the additional bounded trajectories take the form of a saddle connection or a stable cycle to which trajectories starting near the targeted steady state converge. Second, we introduce government debt (see Section IVB). Even with government debt as a predetermined variable, Supplemental Appendix D.1 shows that the determinacy properties of the economy with government debt mirrors that in our baseline economy, except that the economy can no longer jump between the two steady states. Again, depending on how countercyclical risk is, the additional bounded trajectories take the form of a saddle connection or a stable cycle.

¹⁶Bilbiie, Primiceri, and Tambalotti (2023) postulate that the probability that an unconstrained household stays unconstrained is given by $\ln s_t = \ln s_0 + s_1 \ln y_t$. We translate the probability $1 - s_t$ into an arrival rate by using the conversion formula $1 - s_t = 1 - e^{-\lambda_t}$, which can be simplified to yield $\ln s_t = -\bar{\lambda}_t y_t^{-\Theta}$. Imposing steady state $y_t = 1$, where $s_t = s_0$ and $\lambda_t = \bar{\lambda}_t$, we can set $\bar{\lambda}_l = -\ln s_0$ and $d \ln \lambda / d \ln y = s_1 = -\bar{\lambda}_l \Theta$. Bilbiie, Primiceri, and Tambalotti (2023) set $s_0 = 0.987$ and estimate s_1 to lie in the range between 0.2880 and 0.3920, which implies that $\bar{\lambda}_l = 0.0131$ and $\Theta \in [21.98, 29.9]$.

¹⁷A 10 percent decline in consumption is well within the range of empirical estimates; for example, Cochrane (1991) finds that the consumption growth of households who lost their job was 24–27 percent lower than households who did not; Ganong and Noel (2019) find that the consumption of households who become unemployed drops by around 11 percent when unemployment benefits expire, while Gruber (1997) documents that food consumption falls on average by 6.8 percent after households become unemployed.

while $\Theta^* = 31.08$. Thus, the mildly procyclical risk region corresponds to $\Theta \in (0, 15.8)$, the moderately countercyclical region corresponds to $\Theta \in (15.8, 31.08)$ and the highly countercyclical region features $\Theta > 31.08$. Using the estimates of Θ from Bilbiie, Primiceri, and Tambalotti (2023), which lie in the range $\Theta \in (21.98, 29.9)$, with a modal estimate of $\Theta = 28.1$, one can see that these estimates comfortably lie within the *moderately countercyclical* region of the parameter space. The characterization above showed that when risk is moderately countercyclical, in addition to stagnating at the untargeted steady state, the economy can also get stuck in a cycle and remain permanently away from the targeted steady state. In Figure 2, panel B—which is plotted with $\Theta = 28.1$ —the untargeted steady state features a level of output that is about 6.5 percent lower than in the targeted steady state. While this constitutes a considerably large output gap, it is comparable to output-gap estimates in the United States and the euro area following the Great Recession (see, e.g., Summers 2016; Jarociński and Lenza 2018). Furthermore, as discussed earlier, in addition to this large decline in economic activity, self-fulfilling fluctuations can also generate smaller cyclical fluctuations: The stable cycle that surrounds the targeted steady state features output-gap fluctuations with an amplitude of approximately ± 2.5 percent around the targeted steady state, which is comparable in magnitude to cyclical fluctuations in the United States. In fact, Beaudry, Galizia, and Portier (2020) argue that these deterministic boom-bust cycles account for a substantial part of US business cycle fluctuations. Overall, the analysis above underscores the fact that under a standard Taylor rule, our HANK economy with countercyclical risk is susceptible to self-fulfilling fluctuations with nonfundamental fluctuations of plausible magnitudes.

B. Other Policy Rules

While the analysis in Section III is based on an inflation-targeting Taylor rule, our conclusions are robust to other standard monetary policy rules that have been studied in the RANK literature studying local determinacy. This literature has found that compared to purely inflation-targeting rules—rules that allow the policy rates to display inertial behavior or to also respond to output-gap fluctuations—make local determinacy more likely in RANK economies. Supplemental Appendix E.1 studies the determinacy properties of our economy with the monetary policy rule $i_t = \phi_\pi \pi_t + \phi_x x_t$, while Supplemental Appendix E.2 studies the case in which the policy rate exhibits inertial behavior. In particular, it considers a rule $di_t/dt = \alpha(i_t - \bar{r} - \phi_\pi \pi_t)$, where a smaller α implies a more backward-looking rule. These appendices show that, while a large enough ϕ_x and/or a sufficiently backward-looking rule (small enough α) also make *local* determinacy more likely in our HANK economy with countercyclical risk, they cannot eliminate *global* indeterminacy. In fact, Supplemental Appendix E.1 shows that for any finite combination of (ϕ_π, ϕ_x) (no matter how large), the equilibrium is still globally indeterminate in our HANK economy with countercyclical risk. Similarly, Supplemental Appendix E.2 shows that no matter how backward-looking the rule is (however small α is), the equilibrium is globally indeterminate as long as risk is countercyclical.

C. A More General Specification of Countercyclical Risk

In our baseline model, we have modeled countercyclical risk by assuming that the transition rate at which a ξ_h household switches to ξ_l depends on whether output is above or below its level in the targeted steady state: $\lambda_{l,t} = \bar{\lambda}_l \cdot y_t^{-\Theta}$. Supplemental Appendix E.4 shows that our characterization of global indeterminacy in our HANK model with countercyclical risk *does not* rely on this precise functional form. In particular, Supplemental Appendix E.4 considers the case in which the transition rate $\lambda_{l,t}$ is given by¹⁸

$$\lambda_{l,t} = \bar{\lambda}_l \cdot \Lambda(\ln y_t) = \bar{\lambda}_l \cdot \Lambda(\gamma x_t) \geq 0,$$

where $\Lambda(\cdot)$ is any analytic function that takes nonnegative values and is weakly decreasing in x .¹⁹ Furthermore, in order to make our analysis comparable with the baseline model, we also make two additional normalizations. First, we normalize $\Lambda(0) = 1$, so that the targeted steady state is the same as in our baseline model with $y = 1$ ($x = 0$) and $\pi = 0$. This also ensures that the transition rate from ξ_h to ξ_l in the targeted steady state is given by the constant $\bar{\lambda}_l$ like in our baseline model. Second, we parametrize $\Lambda(x)$ such that $\Lambda'(0) = -\Theta < 0$ —that is, the parameter Θ now only captures the cyclical risk *local* to the targeted steady state.

Supplemental Appendix E.4 shows that our results from the baseline results continue to hold under fairly nonrestrictive conditions. In particular, our characterization of global indeterminacy in the form of a stable cycle surrounding the targeted steady state still holds as long as $\Lambda(\cdot)$ is sufficiently convex in x . Intuitively, this means that a 1 percentage point fall in output increases “risk” (the rate at which ξ_h transitions to ξ_l) more than a 1 percentage point increase in output reduces risk. Supplemental Appendix E.4 also shows that the convexity of $\Lambda(\cdot)$ guarantees the existence of the untargeted steady state. However, Supplemental Appendix E.4 shows that the untargeted steady state exists even if $\Lambda(\cdot)$ is *linear*. While the model is obviously stylized, there are a few reasons to think why $\Lambda(\cdot)$ may be convex. The easiest way to see this is to start by considering very large fluctuations in output. Since $\Lambda(\cdot)$ is bounded below by zero, but it is unbounded above, the function must be convex at least as y becomes large. More generally, if we interpret the state ξ_h as employment and ξ_l as unemployment, then $\Lambda(\cdot)$ is proportional to the inflow rate into unemployment. Empirically, this rate *increases sharply* (relative to a simple linear trend) in recessions (i.e., episodes where y falls), but *does not decrease sharply* in expansions (see, e.g., Figure 1 of Crump, Eusepi, Giannoni, and Şahin 2019). Furthermore, the business cycle asymmetries in these labor market flow variables, which we have argued are a proxy for idiosyncratic risk faced by households, are more pronounced than the asymmetries, if any, of log GDP (see, e.g., McKay and Reis 2008). Thus, it is not merely the case that risk is asymmetric over the business cycle because GDP is asymmetric. Rather, if we interpret these cyclicalities in terms

¹⁸This specification implies that the first derivative of $\Lambda(\cdot)$ with respect to y measures how much the transition rate $\lambda_{l,t}$ changes in response to a 1 percent increase in output.

¹⁹ $\Lambda(\cdot)$ must be nonnegative since it is a component of a transition rate. The assumption that the function is also weakly decreasing is meant to capture the idea that risk is countercyclical: Risk is weakly higher when x is lower.

of a functional relation between risk and GDP (both measured relative to trend), risk is a convex, decreasing function of (detrended) log GDP. Finally, the convexity of $\Lambda(\cdot)$ is broadly consistent with the anecdotal observation that recessions are periods when households' perception of idiosyncratic risk is extremely high, but expansions are not periods when idiosyncratic risk is extremely low, at least not to the same extent.

Supplemental Appendix E.4 also shows that the untargeted steady state exists under even more general conditions. Supplemental Appendix E.4 shows that while the convexity of $\Lambda(\cdot)$ is sufficient to guarantee the existence of the untargeted steady state, it also exists even if $\Lambda(\gamma x)$ is *linear* in x . Thus, our characterization of global indeterminacy in our HANK economy with countercyclical risk does not rely on the choice of the precise functional form of $\lambda_{l,t}$ specified in equation (2).

Finally, it is worth pointing out that the conditions that guarantee the existence of the stable cycle and the untargeted steady state *do not* depend on how idiosyncratic risk depends on economic activity far from steady state. This is reassuring because the empirical literature that studies how idiosyncratic risk varies with the business cycle is quite new, and we have only a weak understanding of the connection between the business cycle and idiosyncratic risk. Supplemental Appendix E.4 shows that these conditions only rely on the behavior of $\Lambda(\gamma x)$ local to $x = 0$. In particular, the earlier condition that required that $\Lambda(\cdot)$ be convex for the existence of the stable cycle only needs to be satisfied locally at $x = 0$. In other words, the existence of the cycle only requires that $\Lambda''(0)$ is sufficiently positive and does not require any additional restrictions on the behavior of $\Lambda(\cdot)$ for any $x \neq 0$. Similarly, the existence of the untargeted steady state only requires that risk is countercyclical local to the targeted steady state—that is, if $-\Lambda'(0) = \Theta > 0$.²⁰

IV. Policy Design to Eliminate Self-Fulfilling Fluctuations

Our analysis has shown that standard monetary policy rules cannot guarantee global determinacy in our HANK economy with countercyclical risk. The key force generating this indeterminacy is that in our economy, the natural rate $r^*(x)$ is endogenous and co-moves with output, and this opens the economy up to the possibility of self-fulfilling fluctuations. In other words, since standard monetary policy rules fail to fully account for endogenous fluctuations in the natural rate, they leave the economy susceptible to “*endogenous* demand shocks.” To see why, suppose that ξ_h households believe that the economy is going to enter a recession, and consequently they face a higher probability of transitioning to the ξ_l state. Holding all else constant, this increases households' desired precautionary savings demand, pushing down the natural rate. If monetary policy does not respond sufficiently to this downward movement in $r^*(x)$, the real interest rate is higher than the natural rate $r > r^*$, incentivizing households to reduce spending. This can be seen via IS curve (14), which shows that when $r_t > r_t^*$, output (gap) growth is positive: $\dot{x}$

²⁰ While the existence of the cycle and untargeted steady state is guaranteed by the behavior of $\Lambda(\cdot)$ local to the targeted steady state, the magnitude and periodicity of the stable cycle also depend on the shape of the $\Lambda(\cdot)$ away from $x = 0$. Similarly, how low output is in the untargeted steady state also depends on the actual shape of $\Lambda(\cdot)$ away from $x = 0$.

$= \dot{y}/y > 0$. In other words, given that $r_t > r_t^*$, households reduce their current consumption, and because of nominal rigidities, this results in lower output, rendering the initial belief self-fulfilling. This acts as a negative endogenous demand shock. We now study policy design that can neutralize these endogenous demand shocks.

A. Monetary Policy

As discussed above, standard monetary policy rules are unable to prevent endogenous demand shocks from causing fluctuations in the natural rate $r^*(x)$. A simple monetary policy rule that addresses this shortcoming can be written as

$$(20) \quad i_t = \bar{r} + \phi_\pi \pi_t + \phi_r (r^*(x_t) - \bar{r}).$$

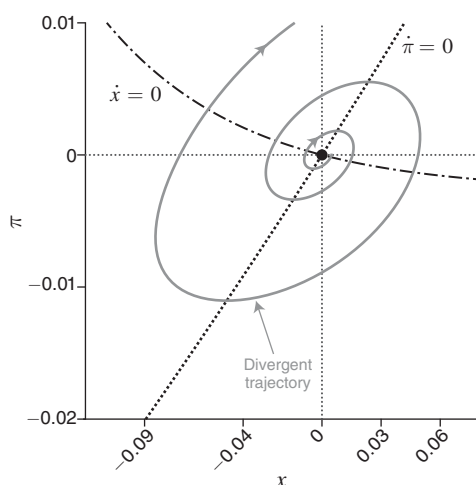
The key change in a policy rule (20) relative to (6) is that monetary policy now also adjusts the nominal rate in response to endogenous demand shocks that result in deviations of the natural rate $r^*(x)$ from its steady state value of $\bar{r}$.

To see how this policy response can eliminate self-fulfilling fluctuations, as before, suppose that ξ_h households believe that the economy is going to enter a recession, and consequently, they face a higher probability of transitioning to the ξ_l state. Holding all else constant, this increases households' desired precautionary savings demand, pushing down the natural rate r^* . However, monetary policy now cuts the nominal rate in response to this lower natural rate. In fact, when $\phi_r \geq 1$, monetary policy lowers the policy rate by at least *one-for-one* with $r^*(x)$. The resulting lower real interest rate undoes the desire to increase precautionary savings, leaving current spending unchanged. Since households do not reduce current consumption, lower output cannot be supported in equilibrium. Hence the initial beliefs about the economy entering a recession cannot be self-fulfilling, and the economy remains at the targeted steady state. Thus, by responding to endogenous fluctuations in $r^*(x)$ sufficiently strongly, monetary policy neutralizes the endogenous demand shock. Analogous to the standard Taylor principle that rules out self-fulfilling fluctuations in RANK, a sufficiently strong response to endogenous fluctuations in $r^*(y)$ discourages self-fulfilling beliefs from taking root in our HANK economy with countercyclical risk. In this sense, it can be thought of as a *Taylor principle, but for natural rates*. This *off-equilibrium* commitment to adjust monetary policy ensures that beliefs about higher or lower output cannot be self-fulfilling. Thus, if monetary policy follows the policy rule (20), even though $r^*(x)$ can deviate from $\bar{r}$, such deviations do not manifest *on equilibrium*. This idea is formalized in Proposition 4.

PROPOSITION 4: *Suppose that monetary policy is described by (20). Then, for any $\Theta > 0$, the targeted equilibrium is globally determinate as long as $\phi_\pi > 1$ and $\phi_r \geq 1$.*

PROOF:

See Supplemental Appendix C.

Panel A. Global determinacy with $\phi_r > 1$ 

Panel B. Stable cycle with unique steady state

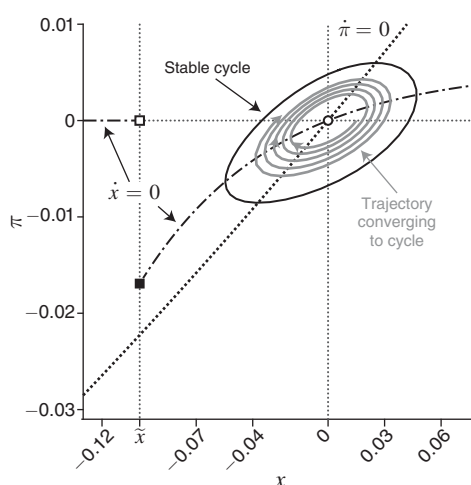


FIGURE 3. PHASE PORTRAITS WITH ACYCLICAL RISK AND COUNTERCYCLICAL RISK

Notes: In both panels, the dotted curve depicts the $\dot{\pi} = 0$ nullcline, while the dash-dotted curve depicts the $\dot{x} = 0$ nullcline. In panel A, the solid gray trajectory depicts a representative trajectory that originates near $(0, 0)$ and then grows unbounded. In panel B, the solid gray trajectory depicts a representative trajectory that originates near $(0, 0)$ and then converges to the stable cycle, which is depicted by the solid black trajectory.

Figure 3, panel A depicts the phase portrait under the monetary policy rule (20) with $\phi_r > 1$. The key difference induced by the commitment of monetary policy to respond to any possible fluctuations in $r^*(x)$ is that, unlike with the standard inflation-targeting rule, the long-run IS curve (the $\dot{x} = 0$ nullcline) is now *downward sloping*. Consequently, the $\dot{x} = 0$ nullcline and the $\dot{\pi} = 0$ nullcline intersect only once at the targeted steady state, implying that this policy rule does not allow the economy to stagnate at an untargeted steady state with below-target inflation and output. Moreover, this commitment to respond to any possible fluctuations in $r^*(x)$ also eliminates the stable cycle, and any trajectory starting away from the targeted equilibrium $(x, \pi) = (0, 0)$ diverges and grows unbounded (as depicted by the solid gray trajectory in Figure 3, panel A).

Endogenous versus Exogenous Demand Shocks.—The reason why the policy rule (20) neutralizes endogenous demand shocks is analogous to the optimal monetary response to *exogenous* demand shocks, which has been studied extensively in the RANK literature. This is easiest to see by setting $\phi_r = 1$ in (20). In this special case, the rule stipulates that holding all else constant, the nominal rate should perfectly track any fluctuations in the natural rate $r^*(x)$:

$$(21) \quad i_t = r^*(x_t) + \phi_\pi \pi_t.$$

This idea is analogous to the optimal response to *exogenous* demand shocks in the RANK literature. As is well known from RANK, exogenous demand shocks cause

fluctuations in the flexible-price real interest rate.²¹ In order to neutralize the effect of these exogenous demand shocks on output and inflation, monetary policy should set the nominal rate to perfectly track the flexible-price real interest rate (Galí 2015). In a similar vein, (21) shows that monetary policy can neutralize *endogenous* demand shocks if the nominal rate tracks the resulting endogenous fluctuations in the natural rate $r^*(x)$.²²

Is Eliminating the Untargeted Steady State Sufficient?—It is worth noting that while the policy rule (20) eliminates both the untargeted steady state and the stable cycle, it is generally not the case that any policy that eliminates the untargeted steady state necessarily eliminates the stable cycle as well. Figure 3, panel B depicts such a case in which monetary policy follows the simple inflation-targeting Taylor rule (6) as long as output and inflation stay *close* enough to the targeted steady state, but switches to a strict inflation-targeting regime, setting $\pi_t = 0$ at all subsequent dates if output falls below some threshold level $\tilde{x}$. This formulation of monetary policy is reminiscent of the commonly studied *escape clause*, where deviations of inflation and output outside a specified range prompt the policymaker to depart from a simple linear rule in favor of a policy that directly restores inflation and output to the monitoring range (see, e.g., Christiano and Takahashi 2018). Figure 3, panel B plots the dynamics under such a rule with $\tilde{x} = -0.1$.²³ Such a policy stance implies that for $x \geq \tilde{x}$, the IS curve is the same as under our baseline, but if $x < \tilde{x}$, the $\dot{x} = 0$ nullcline jumps up to the $\pi_t = 0$ line. Consequently, as long as $\tilde{x} > \underline{x}$ (where $\underline{x}$ denotes the output in the untargeted steady state for a given Θ and ϕ_π in our baseline economy), there is only one intersection between the $\dot{x} = 0$ nullcline and the $\dot{\pi} = 0$ nullcline, implying that such a policy stance eliminates the untargeted steady state from emerging in equilibrium (see Supplemental Appendix C.1 for a proof). However, Figure 3, panel B shows that such a policy stance *does not* rule out global indeterminacy as the stable cycle that surrounds the targeted steady state still exists, and so any trajectory originating in the neighborhood of the targeted steady state initially diverges (since the targeted steady state is locally determinate) but eventually converges to the stable cycle, thus remaining bounded. This shows that simple *escape clauses* (which have often been used as a means of implementing a unique equilibrium), may eliminate some of the multiple equilibria but not all of them. This points to the importance of identifying *all* the forms in which global

²¹ Since our baseline model abstracts from aggregate shocks, the flexible-price real interest rate $\bar{r}_t = \bar{r}$ is constant. More generally, it would change over time in response to exogenous demand shocks—for example, a shock to the discount rate ρ_t . But importantly, even when it is time varying, it does not depend endogenously on the level of output.

²² The specification of monetary policy in (21) is also closely related to the *robust real interest rate rule* of Holden (2024). Holden (2024) shows that a monetary policy rule of the form $i_t = r_t + \phi_\pi \pi_t$, where r_t denotes the *actual* real interest rate, can deliver a determinate equilibrium in a large class of monetary economies. In fact, it would also deliver a globally determinate equilibrium in our HANK economy with countercyclical risk. To see this, notice that using the IS curve (14), the real interest rate in our economy can be written as $r_t = r^*(x_t) + \dot{x}_t$, and so the robust real interest rate rule of Holden (2024) can be rewritten as $i_t = r^*(x_t) + \dot{x}_t + \phi_\pi \pi_t$. This formulation makes it clear that such a rule also delivers global determinacy as it also stipulates that, holding all else constant, the nominal rate should be adjusted to track any potential endogenous fluctuations in the natural rate $r^*(y)$. However, this rule is not exactly the same as (21), since it requires the nominal interest rate to respond not just to off-equilibrium fluctuations in $r^*(x_t)$, but additionally to off-equilibrium fluctuations in $\dot{x}_t$ as well.

²³ $\tilde{x} = -0.1$ implies that monetary policy switches to the strict inflation-targeting regime setting $\pi_t = 0$ at all subsequent dates if output were to decline more than 5 percent below its targeted level.

indeterminacy manifests (as we did in Section III), so that one can design policy that neutralizes all of them.

While at first glance, the specification of monetary policy required to eliminate global indeterminacy might seem abstract, the prescriptions do not constitute a large deviation from the way in which most central banks conduct monetary policy. Following the centrality of inflation expectations for determinacy in the RANK framework, most central banks vigilantly monitor measures of long-run inflation expectations and stand ready to tighten policy if inflation expectations start to drift above the inflation target. In the same way, our HANK economy with countercyclical risk provides a rationale for central banks to pay equal attention to private sector beliefs about real activity. Just as central banks monitor and react to inflation expectations, if measures of confidence in the real economy (such as consumer confidence, households' perceived probability of job loss, etc.) begin to drift down or up, monetary policy should act aggressively to reverse such beliefs. Simply trying to keep inflation expectations on target, while ignoring expectations about real activity, can enable self-fulfilling beliefs that lead to nonfundamental fluctuations in output and inflation.

B. Fiscal Policy

Next, we study whether the global indeterminacy in our HANK economy can be eliminated using fiscal policy. One reason to do so follows from the criticism of the Taylor principle in Cochrane (2011), who argues that the Taylor principle achieves a unique equilibrium by restricting the definition of equilibria to those that feature bounded inflation. Cochrane argues that this restriction, which rules out explosive paths of inflation, is ad hoc and not based on economic theory. Instead, in the context of RANK economies, Cochrane argues that *non-Ricardian* fiscal policy can implement a unique equilibrium without the need for such assumptions. In what follows, we show that such a policy also eliminates the global indeterminacy in our HANK economy with countercyclical risk.

In order to study non-Ricardian fiscal policy, we introduce government debt into our baseline economy. With nonzero debt, the government budget constraint in nominal terms can be written as

$$(22) \quad \dot{B}_t = i_t B_t + P_t(g - T_t),$$

where g denotes government expenditures (we normalize $g = 0$ for simplicity), and T_t denotes lump-sum taxes/transfers. As is standard, we assume that T_t is determined by the fiscal rule:

$$(23) \quad T_t = \bar{r} b^* + \bar{r} \phi_b (b_t - b^*),$$

where $b_t = B_t/P_t$ denotes the stock of outstanding real government debt at date t , and $b^* > 0$ is the level of government debt in the targeted steady state,²⁴ and ϕ_b

²⁴In our formulation, the assumption that $b^* > 0$ ensures that primary surplus is positive in steady state. We choose to focus on the case with positive real interest rates and positive primary surpluses as it distinguishes the

controls how aggressively the fiscal authority raises taxes when debt is above its targeted level. Similar to Leeper (1991), we describe fiscal policy as *passive* when $\phi_b > 1$ and *active* when $\phi_b \in (0, 1)$. As in our baseline model, we assume that monetary policy is still described by the inflation-targeting rule (6). However, unlike the baseline model, we do not restrict $\phi_\pi > 1$. Instead, we require $\phi_\pi \geq 0$, which allows for the possibility that the monetary policy can be *passive* if $\phi_\pi \in [0, 1]$ and *active* if $\phi_\pi > 1$.

Given our assumption of quasi-linear preferences, Supplemental Appendix A.3 shows that we can still summarize household decisions in terms of a single IS curve that is identical to (16a) in the baseline model, and so the dynamics of the output gap, inflation, and (real) government debt can then be described by the IS equation (16a), the Phillips curve (16b), and the government budget constraint (in real terms):

$$(24) \quad \dot{b}_t = (i_t - \pi_t)b_t - T_t,$$

where i_t is given by the inflation-targeting rule (6), and T_t is given by the fiscal rule (23).

Active Monetary, Passive Fiscal.—Simply adding government debt to the baseline model does not eliminate global indeterminacy. As long as fiscal policy is passive ($\phi_b > 1$) and monetary policy is sufficiently active ($\phi_\pi > \varphi(\Theta)$), the determinacy properties of the economy are the same as those of our baseline economy. In fact, Supplemental Appendix D.1 shows that in this setting, if risk is not highly countercyclical $\Theta \in (0, \Theta^*)$ and monetary policy is *active enough*—that is, $\phi_\pi > \varphi(\Theta)$ —then the targeted steady state is still locally determinate as in our baseline model. But, we still have global indeterminacy, and the global dynamics of output and inflation mirror those in our baseline economy. In addition to the targeted steady state, there still exists an untargeted steady state with lower economic activity. But since government debt is a predetermined variable, the economy can no longer simply *jump* between steady states. However, we still have global *indeterminacy*: Supplemental Appendix D.1 shows that when risk is mildly countercyclical, pessimistic beliefs can still cause the economy to slowly transition from near the targeted steady state to the untargeted steady state along a *saddle connection*. Similarly, when risk is moderately countercyclical, self-fulfilling beliefs can still cause the economy to permanently move away from the targeted steady state and to converge over time to a stable cycle that surrounds the targeted steady state.²⁵

source of equilibrium multiplicity in our framework from that in Kaplan, Nikolakoudis, and Violante (2023), who show that in economies with incomplete markets, the fiscal theory of the price level may not yield a unique equilibrium when governments run persistent deficits.

²⁵Furthermore, Supplemental Appendix D.1 also shows that the untargeted steady state is *locally indeterminate*, which implies that in the neighborhood of the untargeted steady state, for a given level of government debt, there are multiple combinations of (x, π) starting from which the economy converges to the untargeted steady state, implying global indeterminacy.

Passive Monetary, Active Fiscal.—In contrast, Supplemental Appendix D.2 shows that a regime of active fiscal policy alongside passive monetary policy *does* eliminate all the manifestations of global indeterminacy that appear in our baseline model (or in the *active monetary, passive fiscal* regime). In particular, an active fiscal, passive monetary regime eliminates the existence of the untargeted steady state, as well as the stable cycle for any $\Theta > 0$ —that is, it prevents households' beliefs about the economy going down such trajectories from becoming self-fulfilling no matter how countercyclical risk is. This is formalized in Proposition 5 below.

PROPOSITION 5 (Global Determinacy with Passive Monetary and Active Fiscal Policy): *Suppose that $\phi_\pi < 1$ and $\phi_b \in [0, 1)$. Then, the economy has a unique steady state with output and inflation at their targeted values and real government debt equal to b^* . Furthermore, the equilibrium is globally determinate: For any given level of government debt b_0 , there exists a unique (x_0, π_0) such that only the trajectory originating at (x_0, π_0, b_0) remains bounded. Furthermore, this trajectory converges to the targeted steady state in which $x = \pi = 0$ and $b = b^*$.*

PROOF:

See Supplemental Appendix D.2

The fact that the passive monetary, active fiscal regime delivers global determinacy in our HANK economy follows from the standard logic of the *fiscal theory of the price level*. When fiscal policy is active ($0 \leq \phi_b < 1$), the fiscal rule (23) along with the government budget constraint (24) implies that even when the gap between outstanding government debt and its targeted level is getting wider, fiscal policy does not raise taxes sufficiently to narrow this gap. Thus, for a given level of nominal government liabilities, the price must adjust to ensure that in equilibrium, the real value of government debt equals the net present value of future primary surpluses. Consequently, the price level is pinned down uniquely to ensure that the government is solvent (Cochrane 2011).

To see how this logic rules out the existence of the untargeted steady state, suppose that as before, ξ_h households start to believe that the economy is about to permanently enter a recession: Output and inflation will remain permanently below their targeted level. Along with the fact that monetary policy is passive, this deflationary path implies that the interest liability of the government is now higher and consequently, the real value of outstanding government debt would increase over time unless the government raises taxes sufficiently. However, with $\phi_b \in [0, 1)$, the fiscal rule (23) implies that taxes do not increase sufficiently, despite the growing level of real government debt. Thus, if pessimistic beliefs were to actually push the economy into a permanent recession, the government would eventually become insolvent, and the government budget constraint would be violated. This prevents such pessimistic beliefs from becoming self-fulfilling, and hence the untargeted steady state cannot exist in a passive monetary, active fiscal regime. Similarly, with active fiscal policy, households' beliefs of the economy being permanently trapped in a cycle around the targeted steady state cannot be confirmed in equilibrium because along the trajectory of output, inflation and real

government debt implied by such a belief would result in the government budget constraint being violated.²⁶

V. Conclusion

We have shown that if risk is even mildly countercyclical, HANK economies can feature self-fulfilling fluctuation under standard monetary policy rules. This is because in HANK economies with countercyclical risk, the natural interest rate is endogenous and co-moves with output, leaving the economy susceptible to endogenous demand shocks. In order to neutralize these endogenous demand shocks, monetary policy needs to commit to adjusting nominal rates one-for-one with any endogenous fluctuations in the natural rate. Doing so implements a unique equilibrium by ensuring that nonfundamental beliefs cannot be self-fulfilling. If doing so is not feasible, a regime with passive monetary policy coupled with an active fiscal regime can also prevent these self-fulfilling beliefs from taking root.

Importantly, since the multiplicity of equilibria does not stem from the presence of the ELB, it can plague the economy even during a tightening cycle. Moreover, our analysis stresses that large and aggressive rate hikes in response to higher inflation *do not* ensure that inflation expectations will remain anchored around its target level. Instead, our framework suggests that, just as central banks monitor and react to inflation expectations, if measures of confidence in the real economy (such as consumer confidence, households' perceived probability of job loss, etc.) begin to drift down or up, monetary policy must act aggressively to reverse such beliefs if it hopes to keep expectations anchored.

Finally, our findings show that local stability analysis can provide a misleading picture regarding the performance of policy rules: In our economy, even when the targeted equilibrium is locally determinate, multiple bounded equilibria exist. This suggests that researchers using HANK models need to be more vigilant regarding the possibility of multiple equilibria. Furthermore, the diverging conclusions based on local versus global determinacy above have important implications regarding the design of monetary policy, which are particularly relevant for the recent post-COVID inflation surge witnessed globally. For example, based on local determinacy, one would conclude that an aggressive enough response of monetary policy to higher inflation (high enough ϕ_π) should ensure that the target equilibrium is determinate, or in other words, that inflation expectations remain anchored around

²⁶Finally, it is also worth commenting on the relation between our paper and a series of papers by Marcus Hagedorn (Hagedorn 2016; Hagedorn, Luo, Manovskii, and Mitman 2019; Hagedorn 2024), which study nominal determinacy in a class of incomplete markets models. These papers exploit the non-Ricardian features of many HANK models that result in a relationship between the real interest rate and the steady state level of government debt. In particular, these papers show that if one specifies fiscal policy as a rule for the level (or more generally, the growth rate) of nominal debt, such a rule delivers a *locally* determinate equilibrium in this class of models. However, such a specification of policy *does not* rule out global indeterminacy in our HANK economy with countercyclical risk. In fact, it cannot even eliminate local indeterminacy in our framework. The reason behind the inability of such policies in affecting the determinacy properties in our HANK model is because in our model, the steady state real interest rate is independent of the level of real government debt owing to the assumption of quasi-linear preferences. The inability of such rules to eliminate equilibrium multiplicity in our economy highlights the fact that multiple equilibria are not driven by the non-Ricardian features of many HANK models that impart a relationship between the steady state real interest rate and the level of real government debt. Consequently, the types of policy rules that eliminate this indeterminacy are conceptually different than those identified in Hagedorn (2016); Hagedorn, Luo, Manovskii, and Mitman (2019); and Hagedorn (2024).

the targeted level of inflation. However, this conclusion would be incorrect since our global analysis reveals that this prescription (no matter how aggressively monetary policy responds to the higher inflation) can still lead to the economy getting trapped in a situation where inflation expectations become permanently unanchored.

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